Arkar Myint Myat

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About Me

Software Engineering and Information Systems student at the University of Oulu with hands-on experience in frontend and backend development, AI/ML, automated testing, and CI/CD. Built practical solutions involving REST APIs, post-quantum cryptography, machine learning, and Transformer architectures. Contributed to a second-place hackathon solution and a Unity game released on Steam, demonstrating software implementation, problem-solving, and teamwork. Experienced in translating research concepts into working prototypes, evaluating algorithms, and documenting technical trade-offs. Looking for Summer 2027 internships.

Education and Training

Bachelor's and Master's Degree Programme in Software Engineering and Information Systems, *University of Oulu* 01/09/2023 – 31/07/2028

Skills

Programming language
Python, Java, C, TypeScript, JavaScript, PHP

Frontend development
React, Angular, HTML, CSS, Bootstrap, Expo

Backend development
FastAPI, REST APIs, Spring Boot, Express.js, Laravel, PostgreSQL, MySQL, MongoDB, Supabase

AI and ML
Deep learning, CNN, LLM integration, NLP, RAG, MCP, Data engineering, Model Optimization, Model Evaluation, Explainable AI, LIME, SHAP, PyTorch, Scikit-Learn, XGBoost, TensorFlow

DevOps and Testing
Git, GitHub, GitHub Actions, CI/CD, GitLab, Jira, Docker, Kubernetes, pytest, Ruff, Security testing

Projects

Kaleva Stake (JunctionXOulu 2026) 23/10/2026 – 25/10/2026
2nd Place – Beyond the Scoreboard Challenge

- Collaborated in a three-member team to develop an AI-powered sports analytics prototype for Kaleva Media.
- Built backend services and Matchstake simulation logic to update team playoff probabilities as match outcomes changed.
- Integrated sports data APIs to support interactive what-if scenarios and live match analysis.
- Implemented AI-generated post-game articles with fact-checking rules to flag unsupported claims.

Demo: https://panokset.vercel.app/
Source code: https://github.com/Arkar-MyintMyat/JunctionXOulu-Hackathon-

Post-Quantum Cryptography (ML-KEM) – Legacy System Modernization Project 07/09/2026 – 21/10/2026

- Led gateway-to-cloud modernization of a legacy IoT system using Python, FastAPI, ML-KEM-768 key establishment, and AES-256-GCM encryption.
- Built a monitoring dashboard, health checks, metrics, and structured logging to track system behavior.
- Set up CI/CD with GitHub Actions and Docker, supported by pytest security tests, static analysis, and dependency scanning.
- Documented the migration strategy and evaluated security and performance trade-offs.

Source Code: https://github.com/AnuR01/softwaredevproject_ML-KEM

Transformer from Scratch Project (English to Burmese translation) 10/08/2026 – 05/09/2026

- Recreated the original Transformer architecture introduced in Attention Is All You Need paper.
- Built an encoder-decoder Transformer in PyTorch, implementing multi-head attention, masking,positional encoding, and model training.
- Prepared 18,082 bilingual sentence pairs with data cleaning and byte-pair encoding for machine translation experiments.
- Compared the model with the PyTorch Transformer baseline using BLEU and chrF, and investigated performance differences.

Source Code: https://github.com/Arkar-MyintMyat/Transformer_Basic

Explainable AI Heart Disease Risk Prediction Project for Bachelor thesis 01/01/2026 – 31/05/2026

- Developed and evaluated a machine learning pipeline for heart disease risk prediction using Logistic Regression and Random Forest.
- Performed data preprocessing, feature scaling, hyperparameter tuning, and model evaluation using accuracy, precision, recall, F1-score, and ROC-AUC.
- Applied SHAP and LIME to explain individual predictions and analyze global feature importance, improving model interpretability.
- Compared predictive performance and explanation consistency, achieving 90.5% recall and 0.950 ROC-AUC with Logistic Regression on the held-out test set.

Thesis: https://oulurepo.oulu.fi/handle/10024/63492

MigraineMSense - AI powered migraine risk prediction mobile app (Junction 2025) 14/11/2025 – 16/11/2025

- Collaborated on an end-to-end ML pipeline using 600 days of synthetic hourly physiological and lifestyle data, applying preprocessing and feature engineering to workload, stress, and heart rate variability (HRV).
- Used a TensorFlow-based LSTM model trained on 24 hour time-series sequences to predict next hour migraine risk, with Integrated Gradients explaining the contribution of individual features.
- Developed a React Native and Expo mobile frontend, integrating an Express.js REST API and Supabase-backed data pipeline to display predicted risk scores, contributing factors, and historical trends.

Demo: https://youtube.com/shorts/_LTbAN0v-o4?is=mmGVEy2u_giFNu3c

King of Cards - Unity Game Development 01/06/2024 – 15/11/2024

- Collaborated in a three-member development team on King of Cards, a Unity game successfully released on Steam.
- Contributed to scene development, scene setup, and implementation tasks to support game functionality and project requirements.
- Performed manual testing, identified bugs, and supported troubleshooting and issue resolution throughout development.
- Assisted in refining gameplay scenes and resolving technical issues to improve game functionality, stability, and the overall player experience before release.

Product: https://store.steampowered.com/app/3299520/King_of_Cards/

Additional Experience

Student Representative at OfferHolder Hub, *University of Oulu* 01/03/2024 – 31/05/2024

- Planned and produced informative videos introducing the University of Oulu, student housing, campus facilities, and everyday student life to prospective international students.
- Communicated practical information through engaging, student-focused content, demonstrating communication, content creation, and audience engagement skills.

Videos: https://youtu.be/lR3sSJ2xzil?is=C-p7nlE9FY6SnRkX

Language Skills

English	**Burmese**	**Finnish**
Fluent	Mother tongue	Basic (A 1.3)